

FREGATA Flying Liferaft

Julian Fung
01709957

Abstract

This report details the project timeline, management strategy, and major design decisions made over the course of the Flying Liferaft Group Design Project. The reasoning behind each decision is detailed and justified, as well as the measures taken to ensure smooth running between separate Functional Area Teams. The market strategy is also detailed to justify the operational requirements of the final product, and alternative markets for expansion are also explored for potential future development.

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1 Introduction

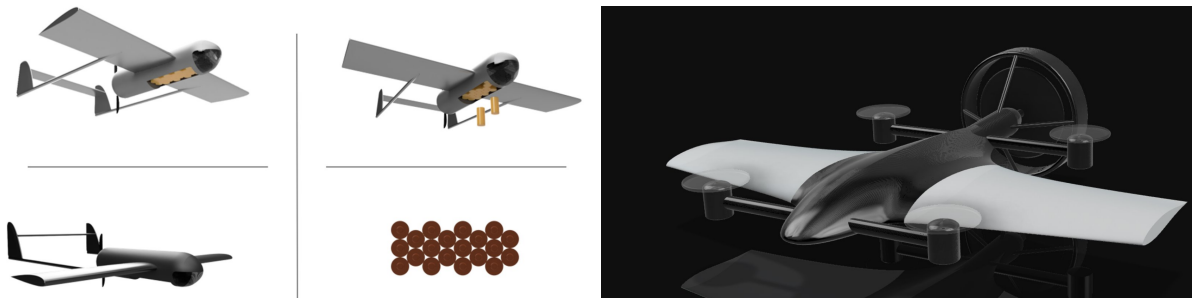
A team of 20 technical specialists were presented with the open-ended yet challenging brief to design an airborne aid solution to people in distress at sea. Split into 5 Functional Area Teams each focused on a different aspect of the design process, effective coordination was required to ensure cohesion between all members. A single design had to be agreed upon, and many executive design decisions had to be made when teams disagreed on different configurations. Furthermore, the marketability of the final product had to be analysed to ensure that there was sufficient demand, and the design constraints resulting from the target market had to be identified and applied. This report documents the process of convergence to the final solution and all major decisions made, as well as the selection process of the target market and the adaptability of the product for other uses.

2 Project Management Strategy and Timeline

Following the initial supervisor kickoff briefing, the project brief was first analysed as a collective to understand the aims and constraints of the task. Following from this, it was decided to split the group into four smaller sub-groups which each contained at least one member from each FAT. The idea behind this initial split was to explore a range of solutions and have each sub-group come up with a rough initial proposal within a week. This way different ideas could be explored and sub-group members would have a larger contribution to the initial ideas such that quieter or less proactive personalities would not be drowned out, and more creative paths could be explored. The proposals were not required to be too detailed, but to have at minimum a general configuration and initial sizing as a proof of concept to show that the design was feasible. Both project supervisors were also included in different sub-groups as it was believed that all members should have an equal contribution to the initial proposals such that no good ideas are missed.

During this initial process, the four sub-groups were left to operate autonomously without much communication between groups as to prevent groups being inspired by others, which could have resulted in very similar proposals invalidating the process. However, it was ensured that literature and resources relevant to the brief were commonly shared as to ensure that all sub-groups had access to any source that was found and deemed useful.

On the Tuesday of week 2, the sub-groups reconvened and each initial proposal was presented and analysed. The resulting proposals are as shown:



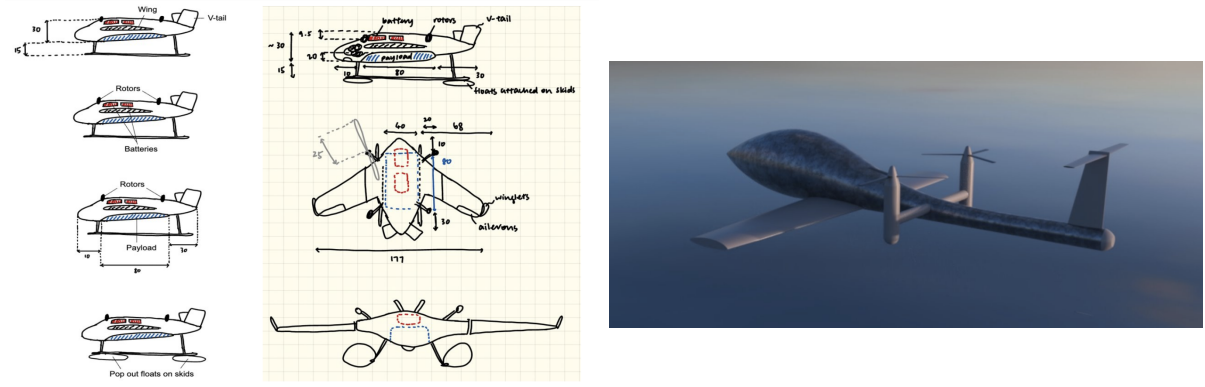


Figure 1: The four initial prototypes

These four initial prototypes covered a wide range of different configurations and design choices, and each had different advantages and disadvantages. A single design was collectively decided upon, the process of which will be explained in greater detail in the next section.

Subsequently, the group was re-divided into functional area teams to begin the detailed design. With a design and configuration decided upon, all the tasks required for the project, such as propeller design due to the selection of a VTOL configuration, could be listed and allocated to the relevant FAT to prevent overlap and ensure that no requirements are missed.

Table 1: The tasks allocated to each FAT

Aerodynamics	Structures	Propulsion & Systems	Performance & Control	Simulation & Testing
Wing Design	Fuselage Design	Powerplant Selection	Mission Performance	Simulation Model
Empennage Design	Material Selection	Powerplant Integration	Point Performance	Testing & Feedback
Aerodynamic Analysis	Wing Structure	Weight & Balance	Stability Analysis	
Fuselage Shaping	Empennage Structure	Systems Design -Electronics -Payload Delivery	Flight Control System -Autopilot -Path Planning	
	CAD Model	Undercarriage Design		

A Gantt Chart was produced to outline the project timeline and task schedule, which can be found in the Appendix. The tasks were allocated as to optimise workflow for each of the FATs and prevent bottlenecking as follows:

In week 2, the Aerodynamics team were assigned to design the wing and empennage as these were the critical components that needed to be sized before performance parameters could be determined. The other FATs were hence assigned to work on tasks that were independent of these parameters, such as fuselage design, powerplant and electronics selection, and the design of the control system framework.

In week 3, the Structures team were hence assigned to design the wing and empennage structure following on from their sizing in week 1. The systems team and stability team were assigned to begin the stability analysis and weight and balance analysis, as these tasks are co-dependent and collaboration would much easier should the teams be working on them simultaneously.

In week 4, following the calculation of the aerodynamic characteristics of the aircraft, the performance, stability, and control team could then analyse the mission and point performance, which could be verified by the simulation and testing team on a rough model. The structures team were allocated to design the secondary structures, which worked in tandem with the propulsion and systems team who could collaborate to design the payload integration mechanism as well as the undercarriage system.

Lastly, week 5 was purely allocated for design iteration to fine tune parameters based on the testing results and feedback of the simulation and testing team.

3 Design Decisions and Conflict Resolution

3.1 General Configuration

The largest design decision made was the general configuration of the final design following the prototyping of the four sub-groups in week 1. To converge to a final idea, the pros and cons of each proposal were considered. To compare the ideas in depth, each sub-group met with each other in turn to critique and draw from one another, adapting ideas to take the best from each. Once all sub-groups had met with one another, the entire group reconvened for an open discussion.

In terms of the aircraft type, it was commonly agreed across the board that a fixed wing configuration was the most effective. This was because fixed wing aircraft have much lower power consumption and faster cruise speeds than their rotary wing counterparts. These two advantages are key for the nature of the problem at hand, as cruise speed needs to be maximised to reach victims faster and maximise survivability. Power consumption is also a huge factor as smaller power requirements lead to smaller and lighter propulsive systems and lower fuel requirements, leaving more space and increasing the maximum payload of the aircraft such that more equipment can be transported to rescue a larger number of victims.

A large source of conflict, however, was the decision to implement VTOL, as three sub-groups designed dual or quad-rotor configurations to allow for VTOL whilst one group designed a pure fixed-wing aircraft to be launched by slingshot. The pros and cons of VTOL were weighed.

Pros:

- The aircraft can be launched and recovered without the need for any infrastructure.
- Contact with saltwater will be minimised, as water landings are not needed, preventing corrosion and damage to moving parts.
- The aircraft is much more maneuverable, and can hover to drop its payload more accurately
- The stall speed can be higher as the aircraft does not need to fly in fixed-wing mode at low speeds

Cons:

- VTOL aircraft are very complex to control

- The rotors required for VTOL will increase drag during fixed-wing flight
- The transition between VTOL and forward flight will be complex to implement and require a tradeoff analysis

In the end, it was decided that VTOL would be implemented due to the fact that a specification point of the brief was that no extra infrastructure was to be used, as well as the fact that the aircraft would be much more reusable due to the fact that it could land directly back on the vessel it was launched from. The lifespan would also be longer due to the reduced effects of corrosion and salt damage, hence further reducing the cost per mission. Furthermore, the increased complexity in control of a VTOL aircraft would not be too large an issue as the control team has the most members out of all of the FATs, and hence the workload could be divided. For this reason we had one member solely responsible for designing the VTOL controller and another solely responsible for designing the VTOL-forward flight transition sequence.

Following the decision to implement VTOL, the propulsion system was decided in unison to be a hybrid between an internal combustion engine as well as electric motors. The internal combustion engine was an idea taken from the prototype designed by sub-group 1, and was chosen to power the pusher propeller during fixed wing flight as petrol is much more energy dense than batteries, reducing the MTOW of the aircraft significantly and increasing its range. However, electric motors were selected for the rotors as controllability is extremely important during VTOL and the control lag of internal combustion engines was an issue raised by the controls team, especially in rough weather.

Due to the unpredictability of incidents at sea, it was decided that a fast turnaround time was a crucial design factor of the aircraft and hence the modular design of the prototype from sub-group 3 was adopted, and it was agreed that a detachable payload module would be implemented to allow for an almost immediate turnaround of the aircraft. The further advantage of having a modular payload is the potential to adapt the aircraft for different uses, which will be explored later in this report.

3.2 Ducted Fan vs Quadcopter Configuration

To implement VTOL capabilities, a conflict arose between two potential configurations. The configurations considered were as follows:

1. A ducted fan at the rear powered by the internal combustion engine, that could produce both downward thrust for VTOL and rearward thrust for forward flight, and two forward rotors.
2. A traditional quadcopter configuration for VTOL with a separate pusher prop for forward flight.

Option 1 was originally preferred as it reduced drag significantly following analysis by the aerodynamics team. However, the controls team raised the issue of the controllability of the rear ducted fan, as internal combustion engines have a significant control lag compared to electric motors. Furthermore, the torque of internal combustion engines varies with rpm, increasing to a peak in the mid-rpm range before decreasing again. This would make the aircraft extremely hard to control, and it would hence need to rely on the two forward rotors for both pitch and roll angle adjustment. Furthermore due to the coupling between pitch and roll angle and translatory motion, this would also have made the

aircraft's movement unpredictable during VTOL, raising safety concerns. Additionally, the propulsion team also raised the issue of the complexity of a ducted fan system that produced both forward and downward thrust, as it would need additional control and more moving parts. This would decrease the reliability and repairability of the aircraft, and also create the need for a larger internal combustion engine to produce additional power especially during the transition phase, increasing its weight. It was hence decided that a traditional quadcopter configuration was more effective and option 2 was selected.

3.3 Battery Charging Method

Due to the decision to implement a hybrid propulsion system, the idea of including an alternator to recharge the batteries was considered. This would have had many advantages as the aircraft could have had more flexibility for VTOL phases as the batteries would be recharged during missions, as well as the removal of the need for manual battery charging. However the sizing of a suitable alternator for the power required revealed that the smallest alternator weighed 9 kilograms, which makes up almost 20% of the aircraft's MTOW. This would have had a detrimental impact on the aircraft's performance, as the payload would have had to be reduced, or the wing resized and the cruise speed reduced. The option of leaving the internal combustion engine running on shore to recharge the batteries was considered, however this would not only have prevented the ability for a rapid turnaround, but also create the need for a clutch system to disconnect the rear pusher prop from the running engine so that it would not keep rotating and cause a safety hazard. Hence it was decided that an alternator was unfeasible for the aircraft and instead, the batteries were designed to be modular and hot-swappable for fast turnaround times, and that they would be charged externally from a standard mains outlet.

3.4 Quadcopter Pylon Configuration

Following the decision for a conventional quadcopter configuration for VTOL, the pylon configuration was considered, and debate arose between having lateral pylons extending outwards from the fuselage and longitudinal pylons extending from the leading and trailing edges. CFD analysis by the aerodynamics team revealed a significant drag reduction with longitudinal pylons due to them being aligned with the freestream, as well as minimal lift reduction from the interaction between the pylons and wings. Further structural FE analysis showed that the main spars would be strong enough to withstand the additional bending moment on the wing structure from the pylons. Following this it was hence decided to implement longitudinal pylons, attaching the quad rotors to the wings.

3.5 Fuselage Diameter

At the end of week 3, the structures team presented a conflict they had within their FAT: two fuselage designs had been devised, one with a diameter of 280mm and length of 1350mm and another with a diameter of 360mm and length of 950mm. The former had a better fineness ratio and better packing density, however would require repacking of the life raft for it to fit within the dimensions of the payload module. As a result, it was decided to progress with the latter design, as the additional operational complexity resulting from repacking a life raft was deemed unfeasible, and a lack of data on the packaging and shape of the liferaft meant that it could not even be confidently decided that repacking of the liferaft was even possible.

3.6 Feedback and Iterative Design

Throughout the design process, many iterations of different components were designed. This was largely due to collaboration and feedback between FATEs, such as the resizing of the fuselage and the redesign of the battery charging systems as previously mentioned. However, weekly feedback from a panel of supervisors was also collected, collated and distributed to the entire team, which also spurred many design decisions. For example the questioning of the product lifespan at the end of week 4 lead to a more in-depth fatigue analysis of all cyclically loaded components, which revealed poor fatigue performance of the rotor blades and a resulting redesign of the root section.

4 Market Strategy and Resulting Constraints

Analysing the project brief, it is observed that the only operational requirements for the aircraft are that it locates and delivers rapid aid to people at sea to maximise their survivability until vessels arrive. This offers a very open-ended range of target markets, and hence market analysis is required to determine constraints such as the required range, size, and cruise speed.

4.1 Deaths at Sea

To determine the target market for the product, the demographic of deaths at sea must first be analysed to determine the people and operational area that has the most need for rapid floatation aid. It is observed that the two largest groups that are victims of drowning at sea are professional seafarers working aboard commercial vessels, and migrants attempting sea crossings. The number of cases for each is summarised:

Table 2: Deaths per year of Migrants and Seafarers [1][2]

Deaths/year by drowning	
Migrants	Seafarers
2268	207.2

It is evident that migrant drownings are much more common than those of commercial seafarers, with their average deaths per year over 10 times higher. This is attributed to the large scale of illegal immigration in overcrowded, poor quality vessels that often fail during crossings, combined with the lack of floatation equipment aboard these vessels and the inability of many migrants to swim. Hence it was decided to design a product aimed at solving the migrant crisis. The data can further be broken down by region:

Table 3: Migrant deaths by region (Since 2014) [1]

Mediterranean	Americas	Europe	West Asia	East Asia
16,584	1,743	884	998	1,437

From the table it is seen that the Mediterranean accounts for 76.6% of all deaths, and, being a very small region, hence has a very high density of cases. It was hence further decided that the Mediterranean be the target area of operation of our aircraft. The geographical distribution of these cases is shown:

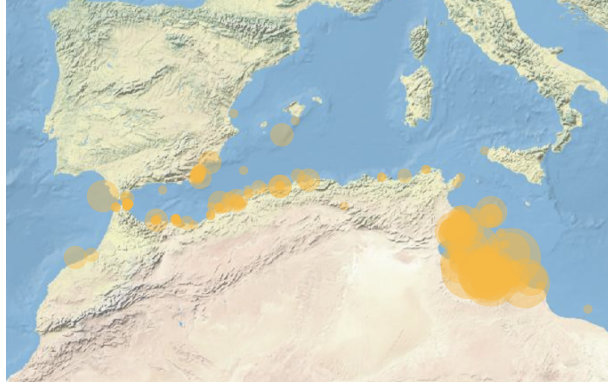


Figure 2: Distribution of migrant drownings in the Mediterranean[1]

The cases are dotted along North Africa, concentrated on the Algerian, Libyan, and Tunisian coastlines, with over 90% of fatalities occurring within 100km off the shoreline[1].

4.2 Range Constraint

It was decided that the aircraft should be launchable off a coast guard vessel to to increase operational flexibility, as these vessels have the mobility to travel between common migration routes. According to the UN conventions on the laws of the seas, coast guard vessels are allowed to enforce customs and immigration laws in the “contiguous zone”, defined as 12-24 nautical miles from the coast. Taking 18 nautical miles, or 33km, as the coast guard vessel distance from shore, and aiming for a service range of 100km, the target range of the aircraft is thus 70km.

4.3 Size Constraint

Due to the decision to launch our aircraft from coast guard vessels, we are hence constrained by the available open deck space for the vertical takeoff and landing. The coast guard services of Algeria, Libya, and Tunisia each operate different patrol vessels, and the constraining free deck space is found to be aboard the Tunisian P350TN, whose largest available free deck area is the poop deck at the stern with a length of 2.5m and width of 4m[3].

From these dimensions, we can hence constrain the wingspan and fuselage length of our aircraft. Maximising the wingspan to increase the aspect ratio and reduce drag, it is determined to align the aircraft with the vessel during takeoff and landing such that the wings span the width of the vessel. Taking a safety margin of 1m either side of the wingtips to allow for the handling of the aircraft by ship crew and also prevent injury from the rotors, the wingspan is hence constrained at 2m.

As the deck is open at the rear, a safety margin of 1m to clear the rotors is taken at the nose of the aircraft only, as the tail will face out to sea. This gives a fuselage length constraint of 1.5m.



Figure 3: The Tunisian P350TN patrol boat[4]

4.4 Cruise Speed Constraint

To find the cruise speed required, the desired response time needed to be determined. With the lowest water temperatures recorded in the Mediterranean being 5°C [5] and the time before the onset of drowning at these temperatures being 30-60 minutes[6], it was determined that a maximum response time of 30 minutes at the maximum range of 70km was required. Accounting for 10 minutes of takeoff, loiter, target identification and payload deployment, we leave 20 minutes of cruise time for maximum range and hence a cruise speed of 58.3 ms^{-1} .

5 Alternative Markets

Although the aircraft is primarily designed to save migrants in the Mediterranean, it can easily be adapted for other uses and markets thanks to its modular payload and camera suite design. Alternative uses for the aircraft can be explored to create more demand for the product, which not only means a larger scale of manufacture, reducing cost, but also additional profit which can be fed back into the research for a potential second generation of the product in the future.

5.1 Private Usage

The most straightforward market to adapt to would be towards private customers including yacht owners, cruise liners and cargo vessels, to rescue passengers or workers that fall overboard. With 27 people falling overboard from cruise ships in 2015 and a mere 17% survival rate[7] due to the immense stopping distance of cruise ships, there is a need for a search and rescue solution. Furthermore the cost of the aircraft is negligible compared to the profit margins and capital cost of yachts and cruise liners.

5.1.1 Private Use Mission Profile

The mission profile of the aircraft for private use would be almost identical to the design mission profile for migrant rescue.

The primary difference in the private use mission profile compared to the mission profile for coast guard use is that the GPS location of the man overboard is known. This reduces the loiter time required and hence increases the range flexibility and endurance of the aircraft. This expands the rescue capabilities to potentially aiding other vessels nearby, should a man overboard call be received by a vessel over VHF carrying the aircraft.

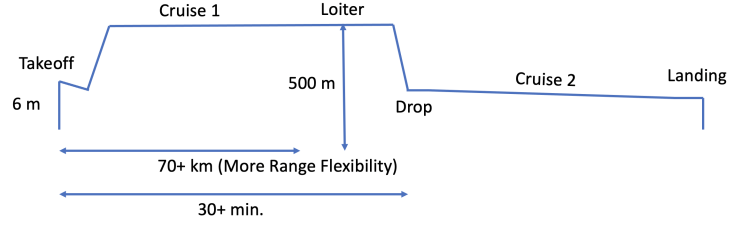


Figure 4: The mission profile for private use

5.2 Press and Media Usage

Another alternative private market is media and press coverage. Currently helicopters are chartered for aerial footage of breaking news, which costs over £600 per hour [8]. By replacing the thermal camera module with a commercial video camera, the aircraft can be used to capture video footage much more rapidly and cheaply than a helicopter, with a mission cost of £150 per hour[9]. Furthermore, as no payload is required, this can be replaced by an auxiliary fuel tank. With a payload volume of 0.0133m^3 , the aircraft can carry 9kg more fuel and the range is increased from 70km to 206.9km (assuming extra fuel is used for cruise).

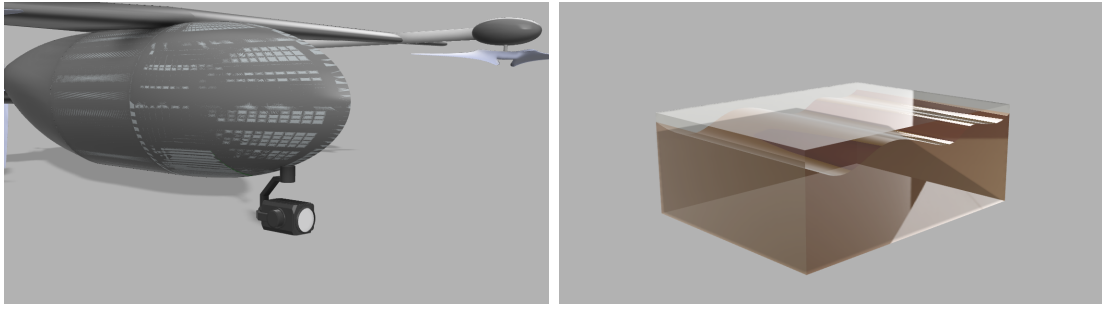


Figure 5: Video Camera Interface and Auxiliary Fuel Tank Module

5.2.1 Media Use Mission Profile

For media use, the main differences are the cruise altitude of 120m, limited by the CAA and FAA, as well as a variable takeoff height depending on the takeoff location. Furthermore an optional loiter and hover is included to capture closer images. Regulations in the UK allow drone flight and filming provided that they are approved by an RAE (Recognised Assessment Entity) and have an approved risk assessment for the mission[10].

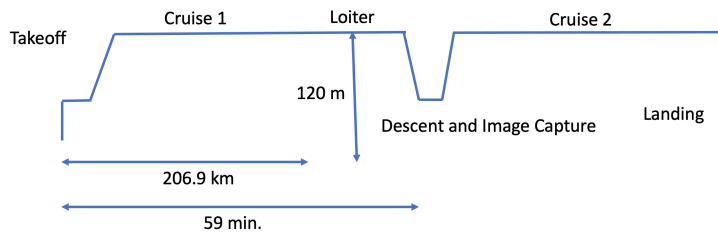


Figure 6: The mission profile for media use

5.3 Police Usage

Along with the Coast Guard, the aircraft can be adapted for use with other civil services, the largest being the police force. It can be used for two primary purposes, namely surveillance/reconnaissance, and crowd control.

5.3.1 Police Surveillance

Currently, aerial surveillance is conducted with police helicopters, which require at least two pilots and costs over £500 per flight hour[11]. By replacing the payload module with an auxiliary fuel tank module similar to the media use configuration, the aircraft can be used for aerial surveillance to aid ground units, taking advantage of its thermal camera to track vehicles or suspects, especially at night when visibility is limited. The auxiliary fuel tank allows for a loiter time of over 5 hours based on the specific fuel consumption of the engine, more than long enough for suspect tracking.

5.3.2 Crowd Control

Crowd control is another potential use for the aircraft, as it can arrive at scenes of civil unrest and remove the need for the deployment of ground units. This can be achieved with a payload module designed to carry crowd dispersal equipment, such as tear gas rounds, to be deployed when needed.

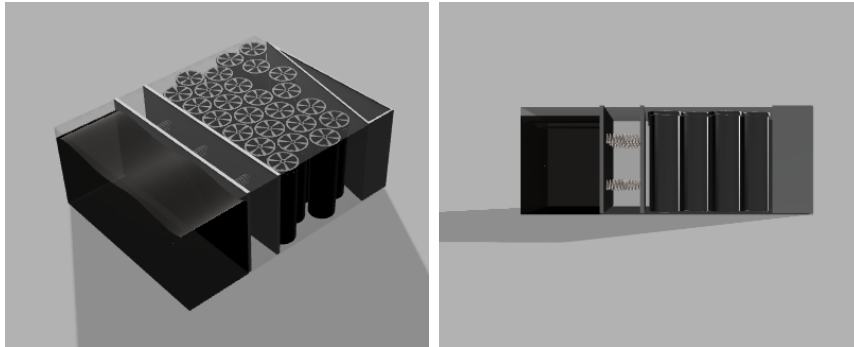


Figure 7: Crowd Control Payload Module

The payload module is designed to carry 25 standard 38mm tear gas shells[12], which are stored within a spring loaded magazine to allow for individual deployment when needed. Due to the additional loiter requirements for crowd surveillance, the module also features a small auxiliary fuel tank at the front which carries an additional 2.65kg of fuel to add 1.5 hours of loiter capability.

When dropping the payload, the loss of weight from each 100g tear gas round causes the weight distribution in the payload module to change.

As can be seen from the graph below, the forwards positioning of the auxiliary fuel tank causes the CG to shift forwards from 143mm to 49mm as the tear gas canisters are dropped if no fuel is burnt, increasing the static margin and hence the static stability of the aircraft. However, if the auxiliary fuel tank is used, the cg shifts backwards slowly to 185mm instead reducing the static margin. Hence the auxiliary fuel tank will not be used until the main tank is empty to maintain the positive static stability of the aircraft.

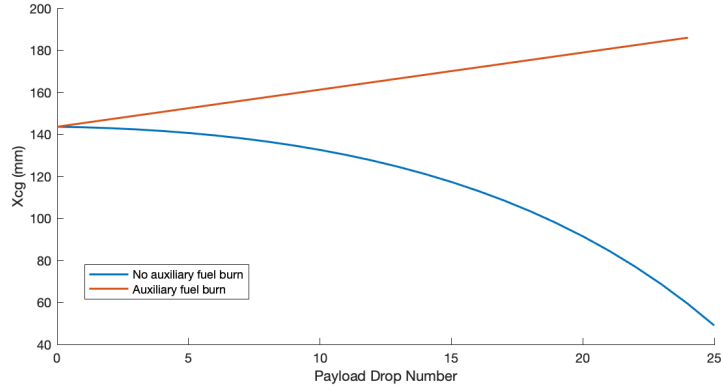


Figure 8: The change in payload CG vs drop no. with and without auxiliary fuel burn

5.3.3 Police Use Mission Profile

The mission profile for the police use is fairly similar to the mission profile for media use, as both use cases assume flight over built-up areas and hence are constrained by the CAA to 120m cruise altitude. However in this case it is assumed that the cruise range will be shorter and the loiter phase longer, with the aid of the smaller auxiliary fuel tank, to allow for surveillance and monitoring before the payload is dropped.

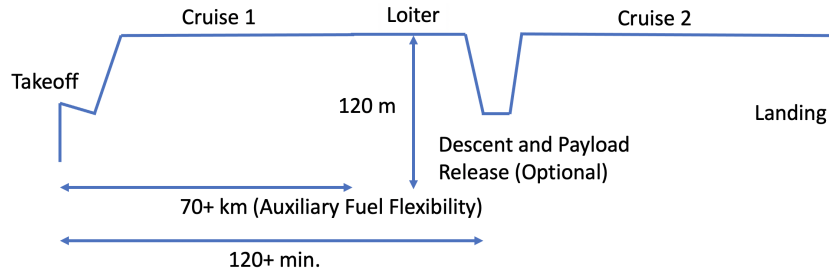


Figure 9: The mission profile for police use

5.3.4 Fire Services Usage

The aircraft can also be similarly adapted for use by the Fire Services. Fires are likely the most urgent emergencies, as they can quickly spread out of control and cause irreversible damage. Wildfires are also a large problem as they are hard to detect, especially in remote areas. The aircraft, with the use of modified payload modules, can carry out both emergency response and also monitor and prevent wildfires from spiraling out of control.

The payload modules designed for fire usage contain extinguishing grenades, spheres designed to activate upon contact with high heat and spread fire retardant powder to cover 1.8 cubic metres[13]. The modules are designed similarly to the police modules, with a spring loaded cartridge system to load the projectiles towards the rear of the module for individual deployment. Depending on the mission required, an auxiliary fuel tank can be included to increase range by 40.35km or loiter time by 1.5 hours, which is useful for wildfire monitoring in tandem with the thermal camera module which can detect high heat levels. If a wildfire is detected, the payload can then be dropped to prevent the spread.

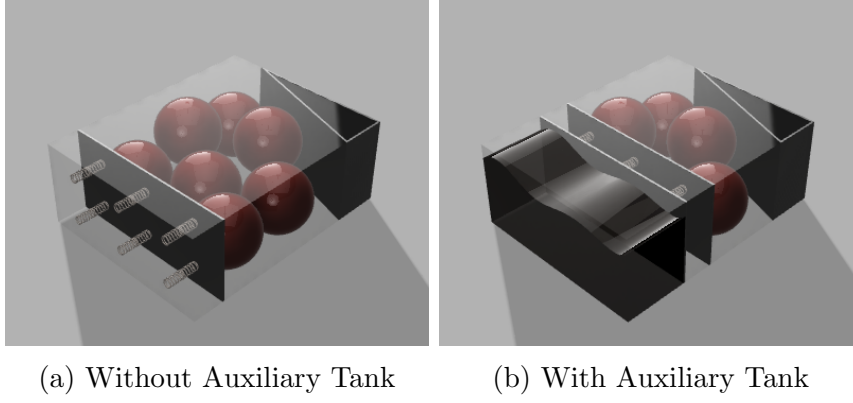


Figure 10: Fire Services Payload Modules

5.4 Fire Services Mission Profile

The mission profile for fire uses varies depending on the payload module installed. Without an auxiliary fuel tank, the aircraft has a range of 70km, whereas with an auxiliary fuel tank it has a range of 110km. However in both cases it cruises at 120m altitude in line with CAA regulations, and has an optional VTOL descent if a payload deployment is needed.

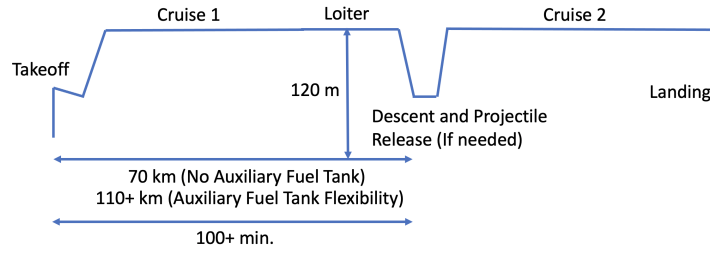


Figure 11: The mission profile for fire use

6 Conclusions

In conclusion the project was a success, with a final design that fulfilled all the specification requirements and targeted an unclaimed gap in the market for sea search and rescue operations between slow, patrol vessel rescue and expensive helicopter missions. The modularity of the design also allows lots of room for expansion, and Fregate has huge potential to be applied for many other uses and markets. There was slight bottlenecking towards the start of the project due to the aerodynamics-heavy analysis required with aerodynamics being one of the smallest FATs, however the project was still delivered on time as no major redesigns were required after initial testing and the aircraft flew stably and passed performance tests. In future, redeployment of members will be used to ensure that projects flow more smoothly.

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Appendix

FLR01 Technical Directors	25-May								31-May
FLR02 Aerodynamic Analysis/Design									
FLR03 Structural Analysis/Design, Manufacturing, CAD									
FLR04 Propulsion and Systems									
FLR05 Performance, Stability and Control									
FLR06 Flight Simulation									
Subgroup 1									
Subgroup 2									
Subgroup 3									
Subgroup 4									

Figure 12: Gantt Chart Week 2

15-Jun	16-Jun	17-Jun	18-Jun	19-Jun	20-Jun
			Design Spec Document, Individual Report DEADLINE		
Design iteration, Report Writing					
Design iteration, CAD, Report Writing					
Design iteration, Report Writing					
Design iteration, Report Writing					
X-P Testing + Feedback, Report Writing					

Figure 15: Gantt Chart Week 5